Copied, Mixed or Computed? Testing Latent Intermediate Values with Minimal Pairs

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Abstract

Probing studies of latent reasoning read a model's state where the answer begins, and take a successful probe as evidence that an intermediate value was computed and a failed one as evidence that it was not. The second fails even for a digit copied from the question, and the first is ambiguous: using prompts that differ in a single token, we show that such a digit is present at that position in two looped and two standard language models, while a probe that sees one prompt at a time reads it at no more than 0.52 even after 45,000 training questions, and every token we varied leaves such a trace, so a probe for a computed value may be reading the inputs. We give a protocol that tests a candidate computed value against copies of the inputs, against generic mixing of them with an operation swap whose bar is set on synthetic mixtures, and for use with a clamp that controls for the copy. We check each test where a value is certainly computed: the clamp detects values the model is about to write or reads four tokens later, and, in small transformers trained from scratch, an unwritten value used eight tokens later; our first, accuracy-based swap failed its positive control, so we replaced it with a version based on shares of variance that passed two fresh controls before we applied it, though in the small transformers it fires wherever the value is present, used or not, and in one of 270 null worlds. In the looped Ouro-1.4B the protocol finds the unwritten intermediate's sign following the operation beyond copies, the answer, the carry and borrow, and generic mixing of the operands, though narrowly and as a small part of the state, which cannot be told apart from the answer with a copied number added back, and no running total shows before that number is read; its use at the position tested is unresolved: a clamp moves the answer in 4% of pairs, against 0.3% for a random one. For a published claim that language models compute sub-answers only while writing a chain of thought, the protocol cannot tell: its operation swap is unclear before the chain, as it is three tokens before a value is written, and the sub-answer's own directions barely move the output, while a test against copies alone and a clamp that leaves the copy in place would both have suggested an early sub-answer.

1. Introduction

Looped transformers apply one block of layers repeatedly (Geiping et al., 2025; Zhu et al., 2025), and one hope is that the extra loops work as a hidden chain of thought; a related question is whether standard models compute sub-answers before they write a chain at all (Kudo et al., 2026). Accuracy alone cannot settle either. Answering directly after the question, Ouro-1.4B's GSM8K accuracy rises from 6.5% after one loop to 23% after four, but almost all of the gain comes at loop 2, where Ouro also improves on benchmarks that involve no arithmetic (Appendix A). To see whether intermediate values are computed, one has to look inside.

Probing studies of latent reasoning read the state at the position where the answer begins, which we call the summary position, and take a successful probe as evidence that a value was computed and a failed one as evidence that it was not (Lu et al., 2025; Yu, 2024; Kudo et al., 2026). The second fails even for the simplest value there is, a digit copied from the question, and the first is ambiguous for the same reason. Our main contribution is a protocol for computed values whose tests are each checked where the answer is known:

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- The problem (Section 3). Minimal pairs, prompts that differ in a single token, show the copied digit at the last prompt token of two looped models (Ouro-1.4B and 2.6B) and two standard ones (Qwen2.5-1.5B and 7B), 0.79–0.88 on average in three models and 0.92–0.97 across Ouro-1.4B's states. A probe that sees one prompt at a time, as probing studies use, reads it at 0.12–0.27 with tens of training templates and at most 0.52 after 45,000 training questions. Every token we varied leaves such a trace, so a probe that reads a "computed" value may be reading the inputs.
- The protocol (Section 4). A candidate value is tested against copies of the inputs, against generic mixing with an operation swap whose bar is set on calibrated synthetic mixtures, and for use with a clamp that controls for the copy. Each step rejects something that a weaker test accepts in our data (Table 2).
- Validation (Section 5). In pre-registered positive controls the copy-controlled clamp detects a value the model is about to write (+0.61 over a random clamp) and one it reads four tokens after writing it (+0.81). Our first, accuracy-based operation swap failed its control, because a code of the operand pair makes every function of the pair readable; a swap based on shares of variance (the share swap), frozen before testing, passed two fresh controls in two model families, with no false positive in 420 null worlds (later, 1 in 270 on another). In small transformers trained from scratch the clamp also detects an unwritten value used eight tokens later, while the share swap fires wherever that value is present.
- Two applications. In Ouro-1.4B (Section 6) the unwritten intermediate $a + b$ is readable just before the answer's last digit, and its sign follows the operation beyond the copies, the answer, the carry and borrow, and a bar set by generic mixing on its own design, though narrowly: on plus problems the sum leads the difference by 0.001 of the variance, its interval just above zero. The effect is small (under 1% of the remaining variance), it cannot be told apart from the answer with c added back, and whether it is used is unresolved: clamping it moves the written digit in 4% of pairs (random 0.3%), and the pre-registered comparison proved weak. For the claim of Kudo et al. (2026) that sub-answers are computed only while the chain is written (Section 7), the protocol cannot tell, with or without a chain: the share swap is UNCLEAR before the chain, as in a control three tokens before a value is written, and b's own directions change the output in at most 3% of pairs; weaker tests would have suggested an early sub-answer.

Most studies were pre-registered, and the text marks the analyses that were not; Appendix F gives the status of every analysis.

2. Setup

Ouro-1.4B and 2.6B (Zhu et al., 2025) run their layers up to four times; we read the state in the middle of loop 1 and at the end of each loop, in fp32 (on check prompts our forward pass matches Ouro's own, with a largest logit difference of 0). Qwen2.5-1.5B and 7B (Qwen Team, 2024) are standard transformers, read after layers 7, 14, 21 and 28 (and 4 in the 1.5B). All four give each digit its own token. Every prompt in Sections 3 and 6 starts with the first four GSM8K training problems (Cobbe et al., 2021), each followed directly by its final answer, then the test question and "Answer:".

The standard probe standardizes the state, projects it onto 256 principal components fitted on the training folds, and fits a multinomial logistic regression whose regularization is chosen by inner cross-validation. It is scored on 5 held-out folds grouped by template, so it must generalize to unseen kinds of problem. The minimal-pair probe is the same probe on states from which each family's mean has been subtracted. Labels are units digits. We probe P0, the last prompt token, and, in Section 6, P1: with the gold two-digit answer fed in, the tens-digit token, after which the units digit is predicted.

3. Why a probe at the summary position is ambiguous

A copied digit is present but hard to read. At the last prompt token a standard probe does not read a digit copied from the question. In a pre-registered study its sanity check, the last digit of the largest number in the question, scored 40.8%, exactly the rate of always guessing the most common digit; by position, a copied digit is read perfectly on its own token and at chance from three tokens on (Appendix B.1). We therefore took one question per template and one of its multi-digit numbers, and made ten copies that differ only in that number's last digit: 137 families from 77 distinct questions. The copies are identical token for token except one, so any difference in a later state is caused by that digit. Counterfactual minimal pairs are a standard interpretability control (Stolfo et al., 2023; Vig et al., 2020); subtracting a family's mean removes what the family shares, the contrast that difference-in-means probing uses to find a direction (Marks &

Tegmark, 2023).

The minimal-pair probe reads the digit on templates it never saw, 0.79–0.88 on average in three models and 0.92–0.97 across Ouro-1.4B's states, while the standard probe reads it at 0.12–0.27 (Table 1, Figure 1). A structured decoder, the best single-prompt decoder we could build (a mass-mean probe from the minimal-pair digit means with a shrunk covariance of the training prompts), does no better (Appendix B.2). The code is shared across questions: in Ouro-1.4B a probe trained on needed numbers reads a number in an added, irrelevant sentence at 0.78–0.97, and a minimal-pair probe of a colour word in that sentence reads it at 0.99–1.00. It is not a magnitude direction (Zhu et al., 2024), since the probe's errors are no closer to the true digit than random guesses, which fits per-digit codes (Levy & Geva, 2025). The minimal-pair probe needs all ten variants of a question, so it tests presence; it is not a decoder for a new prompt.

More questions do not close the gap. In pre-registered studies we gave each model one prompt for each of 6,249 GSM8K training questions, with the units digit of the last number of two or more digits set at random, held out a fixed 1,000, and trained probes on nested subsets of the rest (Figure 3); for the larger models we added 40,000 Orca-Math questions (Mitra et al., 2024), after removing any that reword a test question. At 5,249 training questions the best single-prompt probe reads 0.19 in Qwen2.5-1.5B and 0.36 in Ouro-1.4B; at 45,249 it reads 0.31 in Ouro-2.6B and 0.52 at Qwen2.5-7B's last layer (at most 0.29 at its other layers), against 0.79–0.96 for minimal pairs on the same test questions. The last doubling adds under 0.01 on the GSM8K test questions but still 0.014–0.018 on Orca-Math ones, so data closer to the test questions may read more. The claim is that presence and single-prompt readability come apart by a wide margin at the data sizes tested, not that the digit cannot be read.

Why it is hard to read. The digit's directions are no more crowded than those of a random word swapped in for it or of noise added to its embedding; its change is simply small, and the standard probe reads it about as poorly as the noise (pre-registered; Appendix B.3).

Both kinds of result are ambiguous. A failed probe at a summary position is weak evidence that a value is absent: a digit that is demonstrably present reads below 0.30 in most states. This concerns probes that must generalize to new questions; on prompts that differ only in their numbers, such as bare additions, a standard probe reads both operands at 0.96–1.00 (Appendix B.1). A successful probe is weak evidence of computation: every token we varied, an irrelevant number and a colour word included, leaves a trace at the same position, so a probe that reads a "computed" value may be reading the inputs it is a function of. We did not test whether computed values show the same gap. The protocol addresses the second problem.

4. A protocol for computed values

Within a minimal-pair family one digit varies, so any computed value is a fixed function of it. The protocol therefore varies the operands together in factorial families, with everything else fixed, and asks five questions of a candidate value such as $(a + b) \bmod 5$:

1. Is it more than copies? If the state holds $f(a)+g(b)$, a linear probe on it does no better than the same probe on one-hot codes of the digits, since any function of a digit is linear in its one-hot code. We call that probe's accuracy the copy ceiling, a conditional probe with the inputs as its baseline (Hewitt et al., 2021), and score every label as its excess over its own ceiling. A prime modulus keeps other combinations from sharing the target's parity. Excess shows only that the operands were COMBINED.
2. Does it follow the operation? A generic mixture makes many combinations readable, $a+b$ and $a-b$ alike. Problems therefore come in twin families that differ only in the operation. After regressing the copies (and, where it confounds, the answer) out of the family-centred states, we take the share of the remaining variance that each label's class means explain on held-out families. With D_+ the share of $a+b$ minus that of $a-b$ on plus problems, and D_- the reverse on minus problems, $I = (D_+ + D_-)/2$ cancels any fixed preference for one sign. FOLLOWS requires both Ds positive, I's interval above zero, and I at least 0.30 of N, the needed label's share ($a+b$'s on plus problems and $a-b$'s on minus ones, averaged); DOES-NOT-FOLLOW, I's upper bound below $0.10N$; otherwise UNCLEAR. We call this the share swap. An earlier version compared probe accuracies instead (the accuracy swap); it failed its positive control (Section 5).
3. Is it more than mixing? Synthetic states hold copies of the inputs and one generic nonlinear mixture (products of random codes for pairs of inputs, same-frequency products of periodic digit features, a random network, or a random code for each operand pair, with or without the operation), but no term for the candidate. The bar of 0.30 is 1.5 times the largest ratio among 420 such worlds of varied mixing strength (0.19), rounded up; none of them

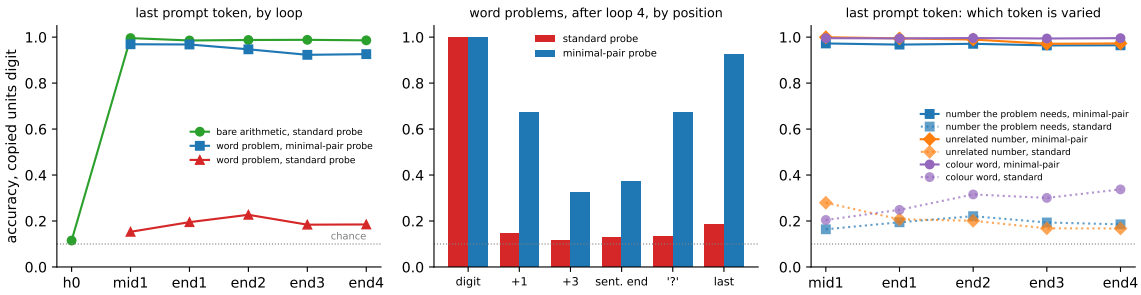


Figure 1. Reading a copied token in Ouro-1.4B (chance 0.10). Left: the last prompt token, by loop. Middle: word problems after loop 4, by position; "+k" is k tokens after the digit. Right: the last prompt token, for a number the problem needs, a number in an added, irrelevant sentence, and a colour word in that sentence. Solid lines: the minimal-pair probe; dotted: the standard probe (right panel: means over two insertion points, so the needed number differs slightly from the per-state values in Appendix B.1).

Table 1. The copied digit at the last prompt token (chance 0.10). Upper rows: 137 families from 77 questions; ranges over Ouro's states and Qwen's layers 7–28, with the mean in parentheses. Lower rows: 1,000 held-out GSM8K test questions; minimal pairs on 100–150 of them; single-prompt probes best over states and over PCA-256 and full-state logistic regression. Pre-registered verdicts are in Appendix B.2. Ouro-1.4B's family study was exploratory. The best single-prompt state is picked on the test set, which favours the single-prompt probe. *Qwen2.5-7B's last layer; at most 0.29 at its other layers.

	Ouro-1.4B (looped)	Ouro-2.6B (looped)	Qwen2.5-1.5B	Qwen2.5-7B
minimal-pair probe	0.92–0.97	0.72–0.92 (0.79)	0.75–0.98 (0.88)	0.58–0.99 (0.85)
standard probe	0.15–0.23	0.14–0.17	0.12–0.14	0.13–0.27
structured decoder	0.17–0.21	0.14–0.17	0.12–0.14	0.14–0.26
minimal pairs on test questions	0.96	0.79	0.91	0.86
single prompt, 5,249 training questions	0.36	0.22	0.19	0.43
single prompt, 45,249 training questions	–	0.31	–	0.52*

passes it, nor any of 420 fresh ones (Section 5).

4. Does it rank first? (Descriptive; not part of the decision.) Compare the target with every combination of the inputs, since periodic number features (Zhou et al., 2024; Kantamneni & Tegmark, 2025) make some combinations easier to read than others.
5. Is it used? Replace the state's component in the candidate's subspace (spanned by family-centred class means, fitted on other families) with that of a partner problem, and read the output. Replacing the whole state is the positive control, and a random subspace the negative one. The subspace is projected off the copy's (and, before an answer, the answer's), since otherwise a clamp can act through the copied digit. USED needs a gain of at least 0.10 over the random clamp with the interval above zero; NOT-USED, an upper bound below 0.02.

Results between a step's bars are UNCLEAR, and a test whose accuracy gate or positive control fails is UNSCORABLE. A computed value needs COMBINED and FOLLOWS, and a used one USED as well. Each step rejects something a weaker version accepts in this paper's data (Table 2); Section 5 tests each step where the answer is known.

5. Validating the protocol

A protocol that never says yes would tell us little, so pre-registered studies tested it where a computed value is certainly used. All but the last use the two-step problems of Kudo et al. (2026), $a = k \pm b$ and $b = d_1 \pm d_2$ with single-digit values, in a short chain format in which the model writes b and then a ("$b = 5$, $a = 6$"); Qwen2.5-7B writes b correctly in 91% of 12,000 prompts. The clamp controls were run after the applications in Sections 6 and 7 had found no USED value; the share swap was validated before it was applied (Table 3).

The clamp. One token before b is written, replacing b's directions, with the copy projected out, makes the model write the partner's b in 61% of pairs, against none for a random subspace (USED). At b's written digit, which the model reads four tokens later to compute $a = k \pm b$, it changes a in 81% (USED; 63% where $a = 0 + b$ copies it). Both results are close to guaranteed, since b is the next token or the token itself. At

Table 2. What each step of the protocol rejects that a weaker test accepts, in this paper's data.

step	weaker test, and what it accepts	§
1. beyond copies	a standard probe: an irrelevant number and a colour word leave a trace at the summary position	3
2. operation swap	excess over copies: every position in the Kudo task is combined (+0.12 to +0.38)	7
2. variance share	accuracy swap: random functions of the pair read 0.99–1.00, so a computed b fails	5
3. calibrated bar	uncalibrated accuracy swap: generic mixing alone reaches $I = +0.14$, against Ouro's +0.157	6
5. copy-controlled clamp	a clamp that keeps the copy moves the written b in 30% of pairs and the answer in 28–35%; with the copy projected out, 2% and 1–3%	7

the end of b's equation, a clamp of the copied digit's directions moves the written b six tokens later in 96% of pairs. Power for a value computed but never written is shown only in the small models below.

The operation swap. The accuracy-based swap failed: one token before b is written, the operands are combined (+0.25) but $I = +0.017$, below its bar, so the protocol did not call b computed. In an exploratory diagnosis the cause was saturation: there, four random functions of the operand pair read at 0.99–1.00, as $a + b$ and $a - b$ do, so accuracy has no room, while the share of residual variance still separates them. We therefore froze the share swap (Section 4), set its bar from 420 synthetic mixing worlds, and judged it only on controls it had never seen: on new problems in Qwen2.5-7B where $a = k \pm b$, and on b in Mistral-7B, it gives FOLLOWS one token before the value is written, and in 420 fresh null worlds it never does. A Qwen2.5-1.5B control failed its accuracy gate (0.78, below 0.80; unscorable) and was replaced by Mistral (after Qwen2.5-3B failed its accuracy pilot) after that result was known, with the statistic already frozen. Three tokens before b is written, and earlier, the same prompts are UNCLEAR and prefer the difference on both kinds of problem ($D_+ < 0$; I/N 0.61 three tokens before), the pattern of the Kudo positions in Section 7; so this pattern does not show that a value is absent.

An unwritten value at a distance. For a value that is computed, used and never written, we trained 6-layer transformers from scratch (5 seeds) on "$x \circ z$; [6 fillers] $w =$" with answer $T[x \circ z] + w$ (mod 23), T a fixed permutation, and negative controls on $T_1[x] + T_2[z] \circ w$, which never need $s = x \circ z$. Distributed alignment search (Geiger et al., 2024), run before any protocol statistic, places s itself (not the operands) at ";", eight tokens before the answer, and at most weakly elsewhere (Appendix E). In the pre-registered test the clamp is USED there in all five models (+0.60 to +0.98 over a random subspace; also with no fillers), and not in any negative control or where s is absent. The share swap is FOLLOWS there in all five, but also at every other position, where nothing uses s but a probe reads it beyond the copies (post hoc), and in one of 270 null worlds, at a tiny needed share, so by its pre-registered rule it is not specific. Here a FOLLOWS marks presence, and only the clamp tests use.

Table 3. Each test where the answer is known (pre-registered). Power: the test says yes where the value is used or computed; false yes: where it is not. Clamps: gain over a random subspace, 1,000 pairs (whole-state patch 0.95 and 0.89 in Qwen). Small models: 6-layer transformers trained from scratch, value eight tokens before the answer (Appendix E).

test: setting	power	false yes
clamp: Qwen-7B, b written next	USED, +0.61	random 0.00
clamp: Qwen-7B, b read 4 tokens on	USED, +0.81	random 0.00
clamp: small models	USED 5/5	none
clamp, Ouro's form: at Ouro's signal	USED 5/5 to share 0.005	–
accuracy swap: Qwen-7B, b	fails	–
share swap: Qwen-7B a, Mistral b	FOLLOWS (0.65, 0.48)	0/420 nulls
share swap: small models	FOLLOWS 5/5	unused positions; 1/270 nulls

6. Application: an unwritten intermediate in a looped model

Studies 1 and 2 used two-step word problems whose answer is $a + b - c$, so the intermediate $a + b$ is never written ("Maya has 30 stickers. Her friend gives her 30 more stickers. She then uses 13 stickers on a card. How many stickers does Maya have left?"), 7,200 problems from 12 templates each. At P1 after loop 4, $a + b$ reads +0.21 (Study 1, mod 10) and +0.22 (Study 2, mod 5) above its copy ceiling, and better than its sign-flipped twin $a - b$ on every one of the 24 templates (post hoc); at P0, before the answer starts, it is not readable. Study 1 missed its pre-registered bar and Study 2's bar was set after it (Appendix C). Every problem needed $a + b$, so a fixed preference for sums would also explain these results. Study 3 swaps the operation.

Study 3: the operation swap. Its 12 new templates come in two versions that differ only in the first step's phrase: "gave her B more stickers" gives $a+b-c$, and "took B of her stickers" gives $a-b-c$. All gates passed (answer-only accuracy 0.67 and 0.77). The accuracy-based swap gives FOLLOWS ($I = +0.157$, Table 4), but that swap later failed its positive control, and generic mixing alone reaches +0.14 on it. Re-scored with the validated share swap (pre-registered after its validation), P1 is still FOLLOWS, with the copies of a, b, c and the answer's digit regressed out: $a+b$ explains 0.006 of the remaining variance and $a-b$ 0.004 on plus problems, and 0.001 and 0.008 on minus problems, against 0.001 for random functions of the operands on average ($I/N = 0.63$, bar 0.30). P1 is the answer's tens digit, which the carry and borrow fix within a family; with them regressed too (pre-registered) it stays FOLLOWS ($I/N = 0.57$; the sum's lead on plus problems falls to 0.001), against a bar of 0.25 from null worlds rebuilt on this design down to Ouro's signal level (none of 270 passes). In the small models, though, one null world of 270 crossed the swap's bar at an eighth of Ouro's needed share, and the swap fired wherever the value was present. P0, before the answer starts, is UNCLEAR. The effect is small: under 1% of the remaining variance, against 11–15% where a value is about to be written (Section 5). No probe label can separate "$a+b$ was formed and the answer computed from it" from "the answer was formed and c added back", since on units digits they are the same function. Nor does training order in 18 small transformers trained from scratch (Appendix C.6). Where c is not yet read, at the end of the sentence that gives b, the share swap is UNCLEAR, though it would likely see a running total as strong as P1's signal, and even the whole state there leaves the answer unchanged (pre-registered; Appendix C.3).

Is it used? For two plus problems from the same family whose units digits of $a+b$ differ but share their parity, we replace x's component at P1 with x''s and read the digit Ouro writes. Patching the answer's subspace after loop 3 moves it to x''s digit in 90% of pairs; patching $a+b$'s does so in 1.7%, less often even than $a-b$'s (3.2%), so the label is NOT-USED (Table 4). An intact answer could override a weak $a+b$ that had been used to form it, so a further pre-registered study clamped the component at the ends of loops 1, 2 and 3, before the answer forms. Clamping the answer moves the written digit in 91% of pairs, and clamping $a+b$ in about 4%, less often than $a-b$: NOT-USED again, also for pairs with equal answers, in which an $a+b$ clamp cannot carry the partner's answer. These interventions predate the copy projection, without which a clamp can also act through the copies. In the small models of Section 5, with its fitting data degraded to Ouro's signal level (share 0.005; $a+b$ about 0.008–0.010) and items per dimension, a clamp of this form still moves the answer in 44–98% of pairs (random 1–12%; USED in 5 of 5, pre-registered; Appendix E). Comparing with $a-b$'s subspace, as Ouro's rule did, is weak there, since it moves the answer about as often. Against random alone (post hoc), the loop-3 patch and the clamp are slightly above chance (+0.017 [0.008, 0.027], +0.039 [0.027, 0.051]), UNCLEAR by the rule, so at P1 the value is readable and its use unresolved.

Table 4. Ouro-1.4B, Study 3, P1 (pre-registered). Upper: excess over the copy ceiling (mod 5) after loop 4, with a 97.5% interval over twin families, the accuracy-based swap and the validated share swap (I as a share of the needed label's, averaged over the middle and end of loops 3 and 4). Lower: share of 2,000 pairs in which the written digit becomes the partner's when the answer's subspace is replaced (positive control), and $a+b$'s hit rate minus the larger of $a-b$'s and a random subspace's, with its pre-registered interval (99.4% and 99.2%). The labels follow the pre-registered rule, whose comparison with $a-b$ proved weak (Appendix E); against the random subspace alone both are UNCLEAR (post hoc).

	plus problems	minus problems
excess of $a+b$	+0.207	+0.078
excess of $a-b$	+0.082	+0.267
accuracy swap I	+0.157 [0.141, 0.172]	
mixing null, largest I	+0.14	
share swap I/N	0.63 (bar 0.30): FOLLOWS	
with carry, borrow	0.57 (bar 0.25): FOLLOWS	
	patch, loop 3	**clamp, loops 1–3**
answer subspace	0.902	0.914
$a+b$ minus control	−0.015	−0.016
	[−0.027, −0.004]	[−0.034, +0.001]
label	NOT-USED	NOT-USED

7. Application: are sub-answers derived before the chain of thought?

Kudo et al. (2026) conclude that language models "have not derived an answer at the moment when input was fed into the model" and compute (sub-)answers while writing the chain of thought. Their evidence is a standard probe at each position and layer, which before the chain reads the sub-answer at 51% and the answer at 18%, against 95–100% inside it and a 90% threshold, and patching. We applied the protocol to their task (Section 5) in two pre-registered studies, in their format with three worked examples, in Qwen2.5-7B. Here a is the answer and b the sub-answer. Families vary d_1 and d_2 over every valid pair, in a plus and a minus version; k varies across families, which

separates b from the answer. We read the end of b's equation and the last prompt token and, without a chain, the token just before the answer digit.

With a chain of thought. Their probe's result is reproduced, at a higher level: before the chain it reads b at 0.64 (their 50.5%) and a at 0.32 (17.9%). The protocol shows what that readability is (Table 5). The operands are combined beyond the copy ceiling at both positions (+0.12 to +0.38), but the share swap does not find it following the operation: it is UNCLEAR at both, with the difference preferred on plus and minus problems alike (on plus problems random functions of the operands explain up to as much as the difference, so this may be a general mix of the digits). A small part does follow the operation (I's interval is clear of zero), but it is faint: in the short chain, where both can be measured in the same prompts, it is about a fifth of b's code one token before b is written ($I = 0.018$ against 0.087, both from the data used to develop the swap). Nor does the chain use such a value. In their format the chain restates b's equation, and replacing the whole state at either position from layer 12 on never changed the b it writes in a 40-pair check before the freeze (not pre-registered). In a short format without the restated equation ("$b = 5$, $a = 6$"), replacing the state at the end of b's equation changes the written b in 96% of pairs, and so does replacing only the copied digit's directions (96%), while b's directions with the copy projected out do so in 2% (+0.021 [0.011, 0.033] over the random clamp; UNCLEAR by the clamp's rule). The pre-registrations made these clamps descriptive, since no position was FOLLOWS.

Without a chain. Asked to answer directly, as in their Implicit setting, Qwen2.5-7B is right on 69% of random problems like theirs, against their 77.8% (76% on our families). Their probe reads b at 0.74 just before the answer (theirs 69.1%). Here the answer is itself a confound, since within a family it fixes b, so the share swap regresses the answer out too and the clamp projects b's directions off both. The operands are again combined at every position (+0.14 to +0.38), but with the answer regressed out as well, the share swap is UNCLEAR at every position, with the same preference for the difference and a small part that follows the operation. Replacing the whole state changes the answer to the partner's in 85–86% of pairs at the two scorable positions (the last prompt token is unscorable). At the end of b's equation the copied digit's directions alone do so in 83%; just before the answer, the answer's directions do so in 46%; b's directions with the copy and the answer projected out, in 1% (NOT-USED) and 3%. The 3% is small but above the random clamp (+0.025 [0.014, 0.039]), so by the clamp's rule it is UNCLEAR (descriptive here too), and with as little as 35% of b's directions left after the projection both tests are weak. Within a family the answer fixes b, so this design cannot separate the two; what we can say is that b's directions beyond the copies and the answer barely matter. Their sentence about failures is not contradicted by a check within each pair of operations (post hoc, Appendix D).

What weaker tests would have said. By its pre-registered rules the protocol cannot tell whether b is held before the chain, in either setting. No position passes the share swap (all UNCLEAR); at three of the five, I/N is above the bar (0.39–0.59) and I (0.006–0.020) larger than in Ouro (0.003–0.004), and only the sign of D_+ keeps them from FOLLOWS. The same pattern appears three tokens before b is written in the control (Section 5), so the swap cannot tell here whether b is held. The interventions find little: b's own directions change the written b or answer in 0–3% of pairs (descriptive; without a chain the clamps keep as little as 35% of b's directions). Two weaker tests would have pointed the other way: the operands are combined well beyond the copy ceiling at every position, and a clamp of b's subspace that leaves the copy in place moves the written b in 30% of pairs and the answer in 28–35%; projecting out the copy (and, without a chain, the answer) removes almost all of that effect, though it also removes a third to two thirds of b's directions.

8. Recommendations

We suggest six practices: test presence with minimal pairs before reading a failed probe as absence; score a candidate against a copy ceiling that includes every input, and the answer and the carry where they fix the token read; swap the operation, and judge the swap on shares of variance against a bar set by synthetic mixtures on the study's own design; do not read an UNCLEAR swap far from where a value is written as absence; read a FOLLOWS as presence, not use; and project the copy out before clamping, with a positive control at the same distance and signal level.

9. Limitations

The probing results rest on four models, and the computation studies on Ouro-1.4B and Qwen2.5-7B, each with one kind of two-step arithmetic read where answers begin. In Ouro, Study 2's bar was set after Study 1's near-miss, several comparisons were run afterwards on saved states, and the signal cannot be told apart from the answer with c added back (Appendix F). The

Table 5. The re-test of Kudo et al. (2026), Qwen2.5-7B. Beyond copies: the needed form's excess over the copy ceiling (layers 12–24). Share swap: I (layers 16–28; with no chain, the answer regressed out too); every position is UNCLEAR, with the difference preferred on both kinds of problem. Short-chain swap values (a) were seen while the swap was developed. Clamp (1,000 pairs): share of pairs in which the model writes the other problem's b (short chain) or answer (no chain) after replacing b's directions with the copy projected out (with no chain, the copy and the answer), the copied digit's directions, or the whole state.

position	chain	beyond copies	share swap I	clamp: b / copy / whole
end of b's equation	theirs	+0.38	+0.020	–
	short	+0.37	+0.018[a]	0.02 / 0.96 / 0.96
last prompt token	theirs	+0.14	+0.010	–
	short	+0.12	+0.007[a]	unscorable (whole 0.00)
end of b's equation	none	+0.38	+0.006	0.01 / 0.83 / 0.86
just before the answer	none	+0.27	+0.015	0.03 / 0.24 / 0.85
last prompt token	none	+0.14	+0.013	unscorable (whole 0.00)

swap's bar rests on five kinds of synthetic mixing; real states may mix differently. The interventions test a few positions and subspaces spanned by class means: a value held elsewhere and read through attention, or in a code outside those subspaces, would be missed. In a pretrained model the clamp's power is shown only for written values read at most six tokens later, in Qwen2.5-7B; for an unwritten value at a distance, only in small models trained from scratch, at Ouro's signal level only by degrading the clamp's fitting data, and not for the form used for Kudo et al. (2026). The share swap was validated in two pretrained models and two steps of one task, with one control replaced after the first failed its accuracy gate, and in the small models it was not specific; Ouro, where it gives the paper's one FOLLOWS, is not among them. The re-test of Kudo et al. (2026) uses one of their nine models and one task level, and its projections keep only 35–64% of b's directions.

10. Related work

Looped models and latent reasoning. The answer-only accuracy of the looped Huginn (Geiping et al., 2025) stays flat across recurrence counts, and a logit lens shows no intermediate values (Lu et al., 2025); an unreviewed repository reports the same for Ouro-1.4B-Thinking (mild-rgb, 2026), Kirin (2026) probes Ouro for whether it will answer correctly. Ouro's state is re-derived within each loop (Wang & Reid, 2026; Huang et al., 2026), and in models pretrained from scratch at equal depth looping brings no reasoning advantage (Schwethelm et al., 2026a). In theory a looped transformer can simulate T chain-of-thought steps in T loops (Saunshi et al., 2025), and models trained to drop written steps can compute intermediates internally (Deng et al., 2024; Hao et al., 2025), though implicit multi-step arithmetic in models trained from scratch relies on shortcuts (Lin et al., 2025). In standard models the first hop of a two-hop question is often resolved latently, the full chain less reliably (Yang et al., 2024; Biran et al., 2024); last-token probes rarely detect intermediate steps in prompted implicit reasoning (Yu, 2024), and sub-answers are computed while the chain is written (Kudo et al., 2026).

Probing and its limits. We build on control tasks and selectivity (Hewitt & Liang, 2019), conditional probing (Hewitt et al., 2021) and difference-in-means probing (Marks & Tegmark, 2023); our learning curves are close in spirit to information-theoretic and minimum-description-length probing (Voita & Titov, 2020; Pimentel et al., 2020). Park et al. (2024) define concept directions from counterfactual pairs, as our minimal pairs do. Probing critiques (Belinkov, 2022) stress that a readable property need not be used (Ravichander et al., 2021; Elazar et al., 2021); a probe can also fail because of how its target is framed (Nanda et al., 2023b). Interchange interventions (Geiger et al., 2021; 2024) and causal mediation with counterfactual inputs (Vig et al., 2020; Stolfo et al., 2023; Hanna et al., 2023) test whether a variable is used; subspace patches can mislead by acting through dormant directions (Makelov et al., 2024), and the copy projection guards against a related risk. Ye et al. (2025) read needed quantities at the question's end in models trained from scratch. To our knowledge, no earlier work reads a copied token at a summary position with minimal pairs while a standard probe fails, or tests a computed value against copies, an operation swap and calibrated mixtures.

Number representations. Numbers are encoded partly linearly (Zhu et al., 2024) and partly by periodic features with periods such as 2, 5 and 10 (Zhou et al., 2024; Kantamneni & Tegmark, 2025; Nanda et al., 2023a). Arithmetic can come from a bag of heuristics (Nikankin et al., 2025), which fits the generic mixing we see, and small transformers trained on addition build explicit carry features (Quirke & Barez, 2024).

Impact Statement

This paper presents work whose goal is to advance the field of machine learning, specifically methods for reading what language models compute internally. Such methods are used to monitor models, and the paper's main point is a caution for that use: a failed probe does not show that a value is absent, and a successful one does not show that it was computed or used. We see no other societal consequences that need discussion here.

Reproducibility

Code, pre-registrations, frozen file hashes, analysis outputs and decision rules for every experiment are in a public repository (https://github.com/Aur-Shalev-Merin/present-but-hard-to-read), including per-item results for the post-hoc comparisons. Every pre-registration was committed, and pushed to a private development repository, before its run or analysis; the public repository lists each freeze commit, when it was pushed, and the time of the first run. Each experiment's document records its predictions, its results, and every change made before data.

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A. Answer-only accuracy by loop

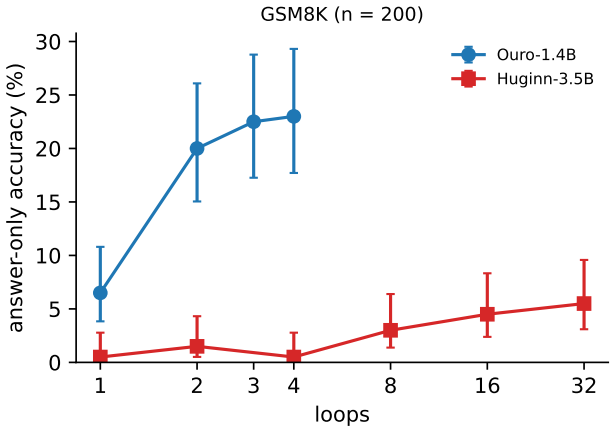


Figure 2. Answer-only GSM8K accuracy by loop, with 95% Wilson intervals.

On the first 200 GSM8K test problems (Figure 2), Ouro answers 6.5%, 20.0%, 22.5% and 23.0% after loops 1 to 4. The gain over four loops is +16.5 points, 95% CI [10.5, 22.5] by paired bootstrap; from loop 2 to loop 4 it is +3.0 [−1.5, +7.5]. Huginn-3.5B reaches 5.5% at 32 recurrences, inside the interval around Lu et al.'s 4.9%, which checks our harness against a published number. We do not require a newline after the answer, because Huginn's tokenizer merges line breaks; an earlier scorer that required one gave Huginn 0%.

The loop-1-to-2 step is not specific to arithmetic. The Ouro paper's own base-model table shows the same shape on every benchmark it reports, including HellaSwag (55.2, 71.2, 74.3 at loops 1, 2, 4) and MMLU (41.2, 60.4, 67.5) (Zhu et al., 2025, Table 10). Ouro's training loss covers the output of every loop (Zhu et al., 2025, Eq. 4), and Schwethelm et al. (2026b) exclude exits after a single loop as consistent with Ouro's exit behaviour, so part of the step may reflect a weak first exit.

On GSM-Symbolic (Mirzadeh et al., 2024), where names and numbers of 100 GSM8K problems are resampled, accuracy on 500 variants rises from 2.8% to 9.2% (+6.4 points [3.4, 10.0], most of it at loop 2), against 10.0% to 18.0% on their 100 originals. A larger set of 4,200 variants from 84 templates has the same shape at a lower level (3.0% to 7.0%). Recall of the published problems is one explanation of the gap, but resampling also changes the numbers' size and difficulty, so the gap is at most a memorization effect. The pre-registered bar for the variant gain was half the GSM8K gain (8.25 points); it fell short and was labelled UNCLEAR. For scale, with generated reasoning Ouro scores about 43% after one loop and 75–77% after two to four (Yang et al., 2026, Fig. 1), and 78.9% after four in the Ouro paper.

B. The copied digit in more detail

B.1. Minimal pairs in Ouro

A standard probe at the last prompt token does not read a digit copied from the question. In a pre-registered study of intermediate values, its sanity check, the last digit of the largest number in the question, scored 40.8%, exactly the rate of always guessing the most common digit. By position, the probe reads a copied digit perfectly on its own token, about half the time one token later (early in the first loop only), and at chance from three tokens on, consistent with the rapid loss of earlier-token information found at the last token of multi-token words in other models (Feucht et al., 2024). In bare questions such as "What is 247 plus 580?" the same probe reads both operands at the last token at 0.96–1.00, but those prompts differ only in their numbers, which makes them close to minimal pairs.

We took one question per template and one of its multi-digit numbers, and made ten copies that differ only in that number's last digit (0–9): 137 such families from 77 distinct questions, one per template (a template's two families vary different numbers of the same question). The copies are identical token for token except one, so any

Table 6. Ouro, the last prompt token (chance 0.10); ranges over mid 1 and the ends of loops 1–4. The trace is the spread of a family's ten variants as a share of the spread between questions.

varied token	minimal-pair probe	standard probe	trace
a number the problem needs	0.92–0.97	0.15–0.23	5–7%
a number in an irrelevant sentence	0.96–1.00	0.14–0.30	3–6%
a colour word in that sentence	0.99–1.00	0.19–0.40	4–6%

difference in a later state is caused by that digit. Counterfactual minimal pairs are a standard interpretability control (Stolfo et al., 2023; Vig et al., 2020); subtracting a family's mean removes what the family shares, the contrast that difference-in-means probing uses to find a direction (Marks & Tegmark, 2023). We then repeated the test with an added sentence that cannot matter ("Unrelated fact: a nearby park has 43 benches."), varying either its number or a colour word ("43 red benches"). Table 6 and Figure 1 give the results in Ouro.

Every varied token reaches the position where the answer begins, and the minimal-pair probe reads it on templates it never saw. Any changed input leaves some change in later states; what matters is that this one is in a code shared across questions, which a probe trained on other templates reads. The code is shared: a probe trained on needed numbers reads irrelevant ones at 0.78–0.97. The irrelevant number almost never changes the first token of Ouro's answer (at most 2.6% of families), so the trace is the copied token, not the answer in progress; this fits GSM-NoOp, where an irrelevant clause lowers accuracy (Mirzadeh et al., 2024), although we did not test whether the trace causes errors. The code is not a magnitude direction (Zhu et al., 2024), consistent with per-digit codes (Levy & Geva, 2025): a linear value component exists (R^2 0.57), but the probe's errors are no closer to the true digit than random guesses (mean distance 3.8 against 3.7), and the top one to four principal components read the digit at only 0.21–0.45, against 0.93–0.97 with 256. It behaves like a code for which token was there, as the colour result suggests. A shared code is what copying predicts, since an attention head that moves a token writes roughly the same vector for it whatever the context; binding vectors that transfer across tasks have been found the same way (Feng & Steinhardt, 2024). The minimal-pair probe needs all ten variants of a question, so it is a test for presence, not a decoder. One sibling does not replace the family: in a pre-registered follow-up, subtracting a single variant with a fixed other digit instead of the family mean reads 0.65–0.90, against 0.79–0.96 with all ten, in all four models of this paper; the gap is largest where the signal is weakest, and it depends on which digit serves as the reference.

B.2. The Qwen replication, and many more questions

A pre-registered study repeated these measurements in Qwen2.5-1.5B with the same families, prompt, probes and code. Its rule called the result a replication if, at the last prompt token over layers 7–28, the minimal-pair probe averaged at least 0.80, while the standard probe stayed at or below 0.30, a structured decoder at or below 0.40, and the gain from half to all training templates at or below 0.05. It is a replication (Table 7). The structured decoder is the best linear decoder we could build for single prompts: a mass-mean probe (Marks & Tegmark, 2023) built from the minimal-pair digit means and a shrunk estimate of the covariance of training prompts around them. It reads no better than the standard probe, and neither does a small MLP, nor, in Ouro, the probe without its PCA step. If the context were Gaussian with the covariance estimated from the training templates, the same decoder would read 0.53–0.92; the gap means that new templates move the state along directions the training templates did not. The claim is about decoders trained without the test question: if the covariance also includes the test question's other variants, a single prompt reads at about 0.9, which amounts to subtracting the family mean.

These data hold only 77 distinct questions, which leaves room for a probe that simply needs more of them. In two further pre-registered studies, one per model, we gave the model one prompt for each of 6,249 GSM8K training questions, with the units digit of the last number of two or more digits set at random, held out a fixed 1,000, and trained probes on nested subsets of the rest (Figure 3). Minimal pairs on 150 (Qwen) and 100 (Ouro) of the held-out questions, with folds by question, read the digit at 0.91 and 0.96. Single prompts stay far below. At 5,249 training questions, 7–39 times the number of components that hold 95% of the states' variance, the best probe, a logistic regression on the full state with its regularization inside the grid, reads 0.19 in Qwen and 0.36 in Ouro; a small MLP reads 0.11–0.14 and 0.16–0.24. Both curves are still rising. By the pre-registered rule

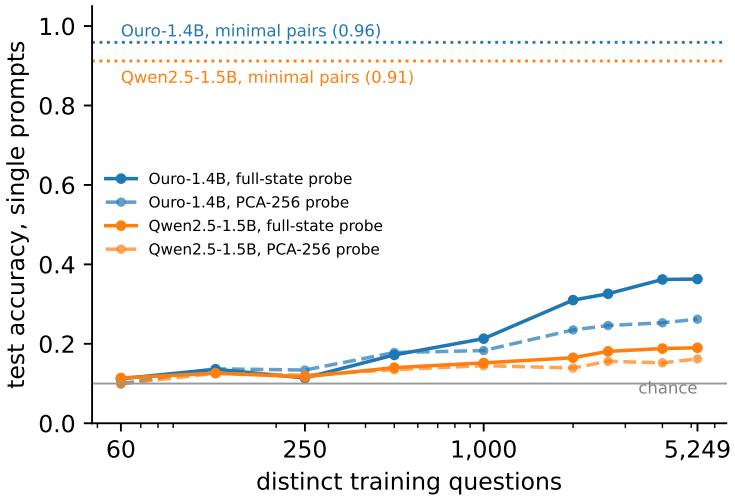


Figure 3. Single-prompt probes trained on more and more distinct questions (last prompt token; best over Qwen2.5-1.5B's layers 7–28 or Ouro-1.4B's loop ends; 1,000 fixed test questions). Full-state: logistic regression on all dimensions; PCA-256: the standard probe. Dotted: minimal pairs on 150 (Qwen) and 100 (Ouro) of the test questions. The best state is picked on the test set, which favours the single-prompt probe.

Table 7. The last prompt token in a looped and a standard model, 77 distinct questions (chance 0.10). Ouro: end-of-loop states, and mid 1 where Appendix B.4 gives it; Qwen: after layers 7, 14, 21 and 28, the pre-registered range. The gain is the standard probe's from half to all training templates.

	Ouro-1.4B (looped)	Qwen2.5-1.5B (standard)
minimal-pair probe	0.92–0.97	0.75–0.98 (mean 0.88)
standard probe	0.15–0.23	0.12–0.14
structured decoder	0.17–0.21	0.12–0.14
MLP probe	0.11–0.17	0.10–0.11
gain, half to all templates	+0.015	+0.004
Gaussian ceiling of the decoder	0.75–0.92	0.53–0.73

Qwen's is partial (+0.044 over the last doubling) and Ouro's climbs (+0.060, above the rule's 0.05, though the level stays below its 0.50). The minimal pairs do not solve the single-prompt task more cheaply: they see all ten variants of the test question, so they show that the digit is present, not what it is in a new prompt. What the gap shows is that presence and single-prompt readability come apart, by a wide margin even after thousands of training questions. Why Ouro's probe climbs faster than Qwen's is open.

A further pre-registered study repeated these measurements in two larger models, Qwen2.5-7B and Ouro-2.6B (Table 1), and extended the many-question curve to 45,249 training questions: the 5,249 GSM8K questions, then 40,000 from Orca-Math (Mitra et al., 2024), after removing any that reword a test question (content-word overlap of 0.3 or more), with 1,000 Orca-Math test questions added to the GSM8K ones. Qwen2.5-7B replicates the 1.5B result by the same rule. Ouro-2.6B shows the same gap, but its minimal-pair mean falls just short of the rule's 0.80, because the digit fades after the first loop (0.92 there, 0.72–0.78 after), so by the rule its result is mixed. With 45,249 questions the single-prompt probes stay well below minimal pairs on the same test questions: 0.31 in Ouro-2.6B, 0.23–0.29 at Qwen2.5-7B's layers 7–21, and 0.52 at its last layer, the one state where a single prompt reads the digit about half the time; that layer already reads 0.43 with the GSM8K questions alone. Averaged over layers and probes, the last doubling adds +0.008 [−0.001, 0.018] in Qwen and +0.004 [−0.006, 0.014] in Ouro on the GSM8K test questions, and still +0.014 and +0.018 on the Orca-Math ones. By the pre-registered rules Qwen's curve is partial and Ouro's is shift-limited: the added questions stopped helping on GSM8K but still help a little on their own distribution, so the result does not say that more data cannot help. Removing the test questions that have a reworded twin among the training questions changes the best probe by at most 0.01.

Table 8. Qwen2.5-1.5B, last prompt token, layers 7–28 (pre-registered, except the input distance, a descriptive check). Input distance: mean distance between the ten variants' input embeddings. Overlap / shuffled: each change's overlap over that of its own shuffled-label subspace. Signal/noise: inside the change's own 9-dimensional subspace. The same ten words and ten noise vectors are used in every family.

change	input distance	overlap / shuffled	signal/noise in subspace	standard probe	minimal-pair probe
a different digit	0.70	0.68–1.19	0.0009–0.0037	0.12–0.14	0.75–0.98
a random word	1.42	0.50–1.06	0.011–0.205	0.24–0.74	0.95–1.00
random noise	1.46	0.75–1.24	0.0001–0.0010	0.11–0.15	0.76–0.99

B.3. A small change, not a crowded one

A trace at a few percent of the variance would still be easy to read if questions hardly varied along its directions. They vary a lot. Take the digit's directions to be the 9-dimensional span of the family-centred digit means, and its overlap to be the context's variance per dimension inside that span over its average per dimension. The overlap is 6.5–11.7 in Ouro and 15–24 in Qwen, so within its own subspace the digit's signal is at most a few percent of the variance around it (Ouro by state in Appendix B.4). The overlap is not specific to the digit. A subspace fitted with the digit labels shuffled within each family, which carries no shared code, has about the same (ratio 0.83 in Ouro and 1.01 in Qwen), and a colour word and a number varied in the added sentence are each about as crowded as their own shuffled-label baselines, or less. Any change within a question lands on the directions along which questions vary most.

What differs between changes is their size (Table 8). In Qwen we also replaced the digit with one of ten random words, or added one of ten random vectors of a typical token embedding's size to its embedding. All three changes land on equally busy directions: by the pre-registered measure, the word swap's overlap is 0.89 times the digit's and the noise's 1.07. Their signal-to-noise inside their own subspaces differs by orders of magnitude, and readability follows it: a standard probe reads the word swap at 0.24–0.74, and the digit and the noise at 0.11–0.15. The input embeddings show where part of the difference starts. In both models the ten digits sit about half as far apart as random words do (0.70 against 1.37 for 60 random words in Qwen, 0.88 against 1.54 in Ouro; the ten swap words of Table 8 are 1.42 apart). But the start does not explain all of it: the word and noise variants start equally far apart, and only the word change reaches the last token, while the noise arrives as small as the digit or smaller. These comparisons are descriptive. A pre-registered follow-up measured the linear response of the last token to small changes at the digit's input embedding (Appendix B.4). To first order, the last token is about as sensitive to random input directions as to directions in which token embeddings differ (gain ratio 1.2–1.5), and the word swap's effect is almost entirely nonlinear: its linear prediction is orthogonal to the actual change and 4.5–15 times too small. What reaches the last token depends on which token is present, not on how far the input moves. Small changes along digit directions are even carried more strongly than along word directions (3–8 times, to first order), so the digit's small trace goes with swapping one digit for another, a smaller change in kind than swapping in a word. Which mechanism carries a token's identity to the last token is open.

The same null, pre-registered for the larger models, gives the same picture in Qwen2.5-7B (word swap 0.90 and noise 0.84 of the digit's overlap; the word swap reads at 0.27–0.93, the digit at 0.13–0.27 and the noise at 0.10–0.12). In Ouro-2.6B it is mixed: the noise is as crowded as the digit (1.28 times its overlap) and as unreadable, but the word swap lands on quieter directions (0.34 times) and reads at 0.82–0.86. There, both size and crowding favour the word, and "small, not crowded" describes the digit against noise only.

This is not superposition in the usual sense (Elhage et al., 2022): the digit shares its directions with the variation between contexts, not with a fixed set of other features. Probing work usually asks whether a probe's success reflects the representation or the probe's capacity (Hewitt & Liang, 2019); here a probe fails because a signal shared across contexts is small next to the variation the contexts add along the same directions, and only the minimal-pair contrast removes that variation. The same limit is known from population coding in neuroscience, where only noise along the signal's own direction limits a decoder (Moreno-Bote et al., 2014).

Table 9. Ouro, last prompt token, held-out templates. Overlap: context variance per dimension inside the digit subspace over its average per dimension; shuffled: the same for a subspace fitted with the digit labels shuffled within each family. Structured decoder: a mass-mean probe with a shrunk covariance, built from the minimal-pair digit means and the training prompts, applied to raw single prompts. Gaussian ceiling: the same decoder on samples from its own fitted model. Standard probe trained on 50% and 100% of the training templates. No PCA: the standard probe on all 2,048 dimensions, C from 10^{-4} to 10. MLP: one hidden layer of 256 units on the 256 principal components.

state	overlap	shuffled	signal/noise in subspace	structured decoder	Gaussian ceiling	standard probe 50% / 100%	no PCA	MLP
mid 1	11.7	10.7	0.005	0.17	0.75	0.14 / 0.15	0.16	0.11
end 1	7.2	9.1	0.014	0.17	0.90	0.18 / 0.19	0.18	0.13
end 2	6.5	7.9	0.023	0.21	0.92	0.21 / 0.23	0.24	0.17
end 3	9.2	10.1	0.011	0.20	0.88	0.17 / 0.18	0.19	0.17
end 4	9.4	11.3	0.010	0.18	0.85	0.17 / 0.19	0.18	0.12

B.4. Ouro by state, and further observations

The minimal-pair probe already reads the digit at 0.87–0.92 from a quarter of the training templates. Against their own shuffled-label baselines, the colour word and the number varied in the added sentence have overlaps of 0.53–0.83 and 0.85–1.13 of the baseline in Ouro. In Qwen both are at about their baseline, except at the final layer, where the colour becomes readable to the standard probe (0.62–0.70) while the digit stays at 0.15–0.18. In a profile over all 29 depths of Qwen at the last token, the minimal-pair probe dips in the middle of the network (0.58–0.63 at layers 18–20) before it recovers to about 0.90. The top within-family residual directions have a higher overlap than the digit's subspace (15–23 in Ouro).

Linear response at the varied token. A pre-registered study took 24 of the questions, one per template, with their original digit and the same prompt, and measured by finite differences how the last prompt token's state responds to small changes of the digit's input embedding (step 0.02 times the median embedding norm). The squared gain along directions in which word embeddings differ is 1.16–1.28 times that along random directions in Qwen (layers 7–28) and 1.36–1.50 in Ouro (loop ends), against the 3 that would have counted as tuned to token directions; we had predicted tuned. Along digit-difference directions it is 2.7–8.5 and 4.4–5.0 times the word-direction gain. The response is linear only locally: for the full word swap, the linear prediction is orthogonal to the actual change ($\cos \approx 0$) and 4.5–15 times too small, while the full-size noise acts roughly linearly (cos 0.38–0.68). Per unit input, a word swap changes the last token 10–25 (Qwen) and 38–373 (Ouro) times more in squared size than the noise, and a digit swap 0.12–0.94 and 0.23–0.36 times as much as a digit-to-word swap. Causal tracing corrupts inputs by adding Gaussian noise to token embeddings (Meng et al., 2022), and Zhang & Nanda (2024) recommend replacing tokens instead; here, noise of a token embedding's size barely reaches the summary position, while a token swap does.

C. Ouro: the computation studies in detail

C.1. Studies 1 and 2

Studies 1 and 2 used two-step word problems whose answer is $a+b-c$, so the intermediate $a+b$ is never written; for example, "Maya has 30 stickers. Her friend gives her 30 more stickers. She then uses 13 stickers on a card. How many stickers does Maya have left?" A study has 12 templates × 3 families × 200 of the 1,000 unit-digit combinations (7,200 problems), plus bare additions ("What is A plus B?") with all 100 unit-digit pairs per family. Study 2 used 12 new templates with new wording, objects and names. Both passed their gates: answer-only accuracy after loop 4 was 0.66 and 0.60, the copied digits read at 0.90 or above, and on bare problems the sum's units digit read 1.00 at P1, where it is Ouro's next prediction. Intervals are 97.5% (Bonferroni over the two hypotheses) and resample families. Table 10 gives the pre-registered results.

Study 1's controls did not include combinations of the answer digit with c; since $a + b =$ answer $+ c$, a state holding the answer and a copy of c could make $a + b$ readable without any intermediate. Study 2 added those controls and moved to mod 5. Before the answer starts, at P0, $a + b$ is not readable in either study; Study 3 returns to P0. In bare addition, the apparent sum before writing is generic mixing: with controls, $a - b$ reads better than $a + b$ (Figure 4, left).

Table 10. Pre-registered results, after loop 4: excess over the copy ceiling. Study 1 was scored mod 10 against $a - b$ and $a + c$; Study 2 mod 5 against $a - b$, $a + c$ and two mixtures of the answer with c.

	Study 1	Study 2 (new templates)
$a + b$ at P1	+0.205 [0.172, 0.236]	+0.219 [0.194, 0.244]
margin over best control	+0.118 [0.094, 0.141] ($a - b$)	+0.075 [0.052, 0.098] (answer $+ 2c$)
bar for the margin	0.15	0.025 (set after Study 1)
label	UNCLEAR	passes; fails Study 1's 0.15
$a + b$ at P0	−0.016	−0.007
bare addition at P0	sum 0.71 vs ceiling 0.10 (no controls)	$a + b$ +0.372 vs $a - b$ +0.463: mixing

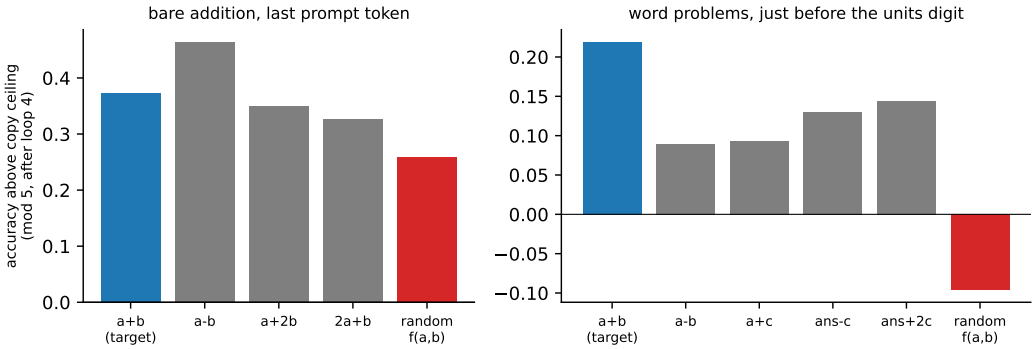


Figure 4. Study 2, after loop 4, mod 5: excess of the minimal-pair probe over each label's copy ceiling. Left: bare addition at the last prompt token. Right: word problems at P1, just before the answer's units digit.

C.2. Sign flips and every combination

The analyses in this subsection were run after the two studies, on their saved states (Appendix F). The sharpest one compares $a + b$ with its sign-flipped twin $a - b$ (Table 11). In both studies $a + b$ reads 0.09–0.13 better after loop 4 and 0.16–0.17 better after loop 3, on every one of the 24 templates. Here the needed sign is plus, so a general preference for differences, which is what bare addition showed, predicts the opposite. For $a - c$ against $a + c$ and $b - c$ against $b + c$ the needed sign is minus, so the same preference predicts the observed winners and the comparison cannot separate the two; $b - c$ also loses to $b + c$ after loop 3.

Against all 30 combinations of the digits other than the answer (which Ouro is about to write, and which reads +0.61 and +0.66), $a + b$ ranks first in Study 2, 0.061 [0.028, 0.093] ahead of the runner-up, and second in Study 1, 0.022 behind it (Figure 5). By the rule we fixed before this check, that weakens the claim that $a + b$ itself is held. The runner-up in both studies is $a - c$. It is another route to $a + b - c$, but it is also the kind of difference a general preference would favour, so we do not count it as evidence of computation.

A further test, used only in Studies 1 and 2, checks mixtures of the output. On units digits, $a + b$ is the answer plus c, so a state that mixes the answer it is about to write with a copy of c could make $a + b$ readable. $a + b$ tops the line answer $+ kc$ by +0.075 [0.052, 0.098] in Study 2 and +0.048 [0.030, 0.066] in Study 1, but a calibrated null on Study 3's items shows that generic mixing tops such a line too (Section C.4); $a + b$ also tops the other two-number combinations by 0.09–0.13. Such mixtures do exist: after loop 4, forms on the answer lines read 0.06–0.07 above forms on no answer line. The margin also survives the stricter baselines. Adding the answer digit already written changes nothing, and adding the carry and borrow, which Ouro does not see as tokens, lowers the Study 2 margin to +0.048 [0.023, 0.074]. On problems Ouro answers correctly it is +0.064 [0.037, 0.091], and resampling templates instead of families does not widen the loop-4 intervals.

No probe label can separate "$a + b$ was formed and the answer computed from it" from "the answer was formed and c added back": they are the same function of the digits. The timing does not help the first reading: after loop 1, $a + b$ and the answer digit read about equally (+0.16 each in Study 2), and after loop 2 the answer reads higher (+0.53 against +0.38). At the moment it is needed, then, $a + b$ is readable beyond what copies allow and beyond its sign-flipped twin, in two studies with different templates, and it tops its answer line by about as

Table 11. Sign-flipped twins at P1 (excess over each label's copy ceiling; difference with 97.5% interval over families). Only the first row has a plus as the needed sign.

needed vs flipped	Study 1, loop 4	Study 2, loop 4	Study 1, loop 3	Study 2, loop 3
$a+b$ vs $a-b$	+0.087 [0.069, 0.105]	+0.130 [0.105, 0.154]	+0.173	+0.161
$a-c$ vs $a+c$	+0.122 [0.093, 0.152]	+0.065 [0.038, 0.092]	+0.064	+0.044
$b-c$ vs $b+c$	+0.052 [0.027, 0.079]	+0.042 [0.015, 0.069]	−0.032	−0.041

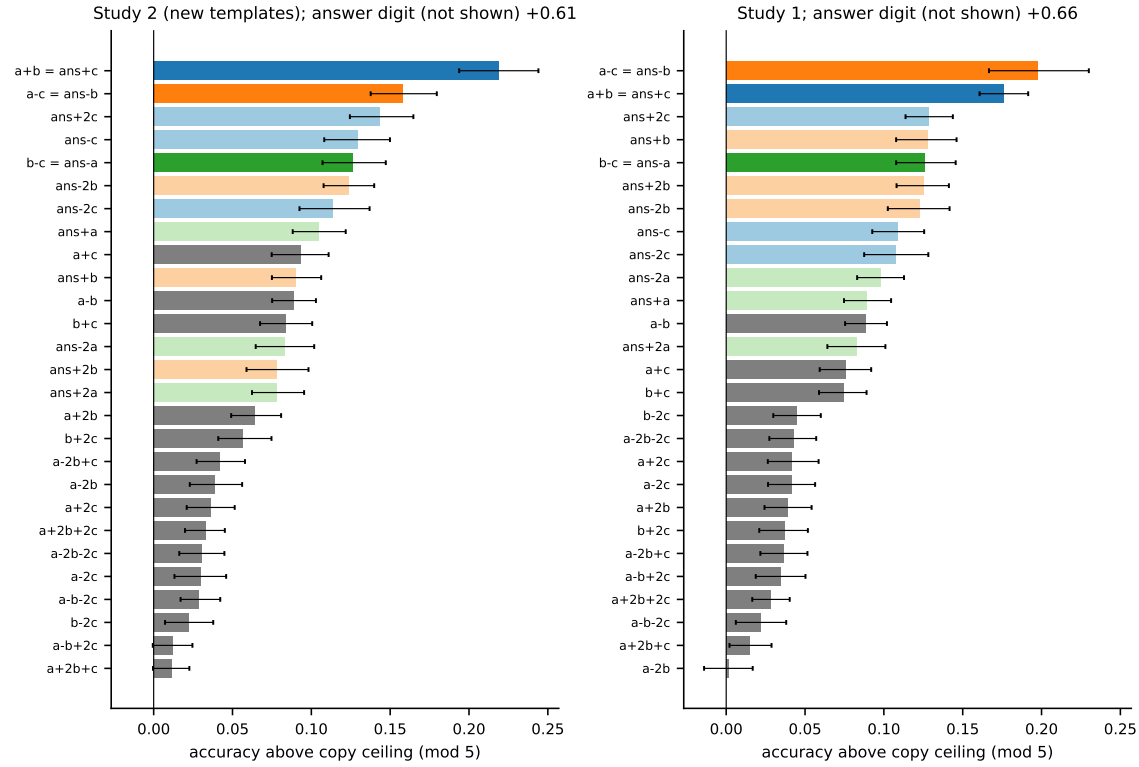


Figure 5. Every mod-5 combination of the operands' units digits at P1, after loop 4: excess over its own copy ceiling, with 97.5% intervals over families. Each colour is one line through the answer (answer $+\,k\times$ one copied number); the dark bar is that line's two-number member ($a+b$, $a-c$, $b-c$). Grey: combinations on no answer line. The answer digit and the copies are not shown.

much as generic mixing does. Two questions remain. Every problem needed $a+b$, so a fixed preference for sums at P1 would also explain the sign-flip result. And whether Ouro uses the value needs an intervention. Study 3 addresses both.

C.3. Study 3: swapping the operation, and patching

Study 3 was pre-registered in two parts: Part A, the operation swap, and Part B, patching. Its 12 new templates come in two versions that differ only in the first step's phrase: "gave her B more stickers" gives $a+b-c$, and "took B of her stickers" gives $a-b-c$. A plus family and its minus twin share names, tens digits and unit-digit combinations. Each version has 7,200 problems, and probes are trained and scored within a version. Let D_+ be the excess of $a+b$ minus that of $a-b$ on plus problems, D_- the reverse on minus problems, and $I=(D_++D_-)/2$. A fixed preference for either sign adds the same amount to one D and takes it from the other, so it cancels in I; we report it as $F=(D_--D_+)/2$. The label FOLLOWS required $I\geq 0.05$ at P1 after loop 4, with its interval above zero and both Ds positive.

All gates passed: accuracy after loop 4 was 0.67 on plus and 0.77 on minus problems, the copies at P0 read at 0.98–1.00 at their best state, and the bare sum digit at 1.00. The label is FOLLOWS (Table 12): $I=+0.157$

Table 12. Study 3, operation swap (pre-registered): P1 after loop 4, excess over the copy ceiling (mod 5), with a 97.5% interval over twin families.

	plus problems (need $a+b$)	minus problems (need $a-b$)
excess of $a+b$	+0.207	+0.078
excess of $a-b$	+0.082	+0.267
needed minus flipped	$D_+ = +0.124$	$D_- = +0.189$
$I = +0.157$ [0.141, 0.172]: FOLLOWS; fixed preference $F = +0.032$ [0.012, 0.053]		

Table 13. Study 3, patching at P1 on plus problems (pre-registered; the loop-2 point and the equal-answer pairs were added before analysis), 2,000 pairs per row. A hit is the digit the patch implies. Same-c pairs move the answer with $a+b$; equal-answer pairs hold it fixed and have no answer patch. Statistic: hit($a+b$) minus the larger of hit($a-b$) and hit(random), with a 99.375% interval.

		hit rate by patched subspace					
pairs	patched after	answer	$a+b$	$a-b$	random	statistic	label
same c	loop 3	0.902	0.017	0.032	0.001	−0.015 [−0.027, −0.004]	NOT-USED
same c	loop 2	0.331	0.007	0.006	0.002	+0.001 [−0.005, +0.007]	NOT-USED
equal answer	loop 3	–	0.003	0.002	0.000	+0.001 [−0.003, +0.007]	NOT-USED
equal answer	loop 2	–	0.003	0.003	0.001	+0.001 [−0.002, +0.003]	NOT-USED

[0.141, 0.172], with 97.5% intervals that resample twin families together. On minus problems $a-b$ reads +0.27 and $a+b$ +0.08, nearly the mirror image of plus problems. The fixed preference is small and favours differences, $F = +0.032$ [0.012, 0.053], as bare addition suggested. The digit Ouro is about to write depends on the carry or borrow from the units, which is sign-specific, so a pre-registered stricter check adds the carry and borrow to the copy ceiling. With them, I at P1 is +0.126 [0.107, 0.145], still well above the bar, and at P0, before the answer starts, it is −0.013 [−0.028, 0.002]: the accuracy swap finds nothing sign-specific there beyond the carry and borrow, and the share swap is UNCLEAR there. The copy of b reads only 0.77–0.81 there after loop 4 (0.66 in Study 1), so P0 is the weaker test. In both versions the needed form also tops the other forms on its answer line after loop 4, by +0.036 and +0.045, which generic mixing matches (Section C.4). So the sign-flip result of Studies 1 and 2 is not a preference for sums at P1: the better-read sign is the one the problem needs. This still does not separate a first step formed before the answer from the answer with c added back, since in both versions the needed form is the answer plus c. These are the accuracy-based swap's numbers; the share swap's re-score of P1 is in Section 6 (FOLLOWS, $I/N = 0.63$, and 0.57 with the carry and borrow regressed, against a bar of 0.25 set on this design).

Part B asks whether Ouro uses the value. For two plus problems x and x' from the same family, we run both to a patch point, replace x's component in a 9-dimensional subspace U at P1 with x''s, $h \leftarrow h + UU^\top(h_{x'} - h)$, run the remaining loops, and read the digit predicted at P1. Each subspace is spanned by the family-centred class means of one units digit ($a+b$, $a-b$ or the answer) at the patch point, fitted on the other half of the templates; a random subspace is a further control. The two problems share c and have different units digits of $a+b$ with the same parity, so a parity code alone cannot produce a hit, and Ouro answers x correctly. A hit is x''s answer digit, the one x would write if its $a+b$ were x''s, and the answer patch is the positive control. The statistic is hit($a+b$) minus the larger of hit($a-b$) and hit(random): USED at 0.10 or more with the interval above zero, NOT-USED if the interval's upper end is below 0.02, and unscorable if the answer patch hits less than 30% of the time. The frozen plan patched at the end of loop 3. After the states were extracted, but before any analysis, we added a patch point at the end of loop 2 and a second kind of pair with equal answers (c shifts to compensate). For those pairs a hit is the digit x would write from x''s $a+b$ and its own c, and a state that holds only the answer with c added back predicts no hits. Each of the four tests has a 99.375% interval (Bonferroni: 2.5% for Part A's test, 0.625% for each patching test).

All four tests are NOT-USED (Table 13). At the end of loop 3 the answer patch moves the prediction to x''s digit in 90% of pairs, so a 9-dimensional patch at P1 can steer the output. The $a+b$ patch does so in 1.7%, slightly less often than the $a-b$ patch, and under it Ouro keeps its own digit in 96–99% of pairs. With the answer held fixed, the $a+b$ patch hits 0.3%. The end-of-loop-2 tests are weak: there the answer patch hits only 33%, because

Table 14. Patching $a+b$ at P1 without a formed answer to override it (pre-registered), 2,000 pairs of each kind. Statistic: hit($a+b$) minus the larger of hit($a-b$) and hit(random), with a 99.17% interval (six tests). In the ablation test the patch is restricted to directions outside the answer subspace; that test has no positive control, and the ablation alone changes the digit in about 68% of pairs.

test	answer patch hit	same-c pairs	equal-answer pairs
patch at end of loop 1	0.009	UNSCORABLE	UNSCORABLE
clamp at ends of loops 1–3	0.914	−0.016 [−0.034, +0.001]	+0.005 [−0.001, +0.012]
answer ablated, end of loop 3	–	+0.005 [−0.011, +0.015]	−0.001 [−0.016, +0.013]

loops 3 and 4 restore x's own digit in about two thirds of pairs, which fits a state that is re-derived in each loop (Wang & Reid, 2026). Together the two parts say that at P1 the readable combination follows the operation, but the patches show no clear sign that Ouro reads it there to write the units digit (against the random subspace alone, post hoc, the loop-3 patch is +0.017 [0.008, 0.027], UNCLEAR). By the end of loop 3, though, the answer is strongly readable, so an intact answer could override a weak $a+b$ component even if $a+b$ had been used to form it earlier; Section C.4 tests this. Only P1 was patched. A pre-registered follow-up on the same problems looked earlier, at the period ending the sentence that gives b ("Her friend gave her 13 more stickers."), where a running total would sit. There the sign did not follow the operation under the accuracy-based swap, which was not re-scored (I at most +0.018 at any loop end, every upper bound at most 0.036; UNCLEAR by its rule, since no bound fell below 0.02), so its patching step was not run, and the answer was not readable at the end of the sentence that gives c either. A later pre-registered study re-scored that period with the share swap, with the copies of a and b regressed and a bar (0.35) set on 270 null worlds built on the position's own design, none of which passes it: UNCLEAR, with N 0.002 and I +0.000 [0.000, 0.001], the difference preferred on both kinds of problem. Planted in null worlds at the mixing found there, a running total is FOLLOWS once its I reaches 0.002–0.0045, half to just over Ouro's at P1, so one as strong as P1's signal would probably have been seen; a fainter one is not excluded. Its clamp over the sentence's tail, with the copies of a and b projected out, is NOT-USED (−0.001 [−0.002, 0.000] against a random subspace, 2,000 pairs), but so is replacing the whole state at every tail token (0.1%), while replacing b's own token moves the answer in 91% of pairs. The answer reads b from its own token, not from the sentence's end, so this test cannot show whether a running total is formed there.

C.4. Patching before the answer forms, and a null for mixing

Two further pre-registered studies address the weakest points above. The first repeats the patching on the same pairs in three ways that leave no formed answer to override $a+b$ (Table 14). A single patch at the end of loop 1, where $a+b$ and the answer read about equally, cannot be scored: loops 2–4 undo any patch made there, even one of the answer (hit 0.009). Clamping, which sets the component to the partner's at the end of loops 1, 2 and 3, does not have that problem: clamping the answer moves the written digit in 91% of pairs, while clamping $a+b$ moves it to the partner's digit in about 4% of pairs, below the $a-b$ control and above the random one (0.3%), and Ouro keeps its own digit in 90–97% of pairs. Both clamp tests are NOT-USED, including the equal-answer pairs, in which an $a+b$ patch cannot carry the partner's answer. In the third test the answer's component is first set to its family mean at the end of loop 3. Ouro then writes its own digit in only 31–32% of pairs, so the $a+b$ component left at P1 does not rebuild the answer, and the partner's does not steer the digit: every condition, random included, hits the target at about chance (0.08–0.10). This test is weaker than the clamp. It has no positive control, and with the answer removed the output is largely scrambled, so a null there may reflect a disrupted state rather than an unused value.

The second study calibrates the protocol's mixture tests. We built synthetic states on Study 3's items with copies of a, b and c, a strong answer code and no $a\pm b$ term, plus one generic mixture of the four digits: products of random codes for every pair, or same-frequency products of periodic digit features, or a random one-layer network. The mixing strength was set so that the needed form reads about as well as in Ouro, and the paper's own probes then scored each world. Pairwise mixing tops the answer line by as much as Ouro does or more (mean margins +0.078 and +0.051 over six worlds each, against +0.036 and +0.045), so by the pre-registered rule the line test does not separate a held $a\pm b$ from mixing. The reason is that the needed form is a function of two inputs, a and b, which pairwise mixing makes readable directly, while the other forms on its line need all

three; in the Fourier worlds, answer $\pm 2c$ barely reads, as same-frequency products predict. The random network never made the needed form as readable as in Ouro, so it could not be calibrated. The same worlds give an operation-swap statistic I of +0.09 on average and at most +0.14, against +0.157 in Ouro: the sign-following signal exceeds every null world, but mixing alone produces about half of it. In small transformers trained from scratch on the same task, dense checkpoints did not separate the two accounts either: the sign-specific signal never appeared while the answer was absent, and two of three architectures solved the two-digit task with no readable first partial result (Appendix C.6). These comparisons use the accuracy-based swap; the share swap's bar comes from 420 calibration worlds instead, and 420 fresh ones gave no false positive (Section 5); 270 worlds rebuilt on Study 3's design set Ouro's bar at 0.25 (Section 6).

C.5. Loop dynamics

At P1, every combination of the numbers we checked reads better at the end of a loop than in its middle, including $a + c$, which the problem never needs, while the copy of b reads better in the middle. This is not an effect of normalization: Ouro normalizes the state it feeds into the next loop, but dividing each mid-loop state by its own size leaves the pattern unchanged. Within each loop, the second half of the layers makes combinations of the numbers readable. Unneeded ones then fade: $a + c$ peaks after loop 2 (+0.46) and falls to +0.09 by loop 4, while $a + b$ peaks after loop 3 (+0.45) and keeps +0.22. In Study 3 the needed form peaks after loop 3 on plus problems (+0.51) and after loop 2 on minus problems (+0.42).

C.6. Training order in small models

A pre-registered study asked whether a sign-specific $a \pm b$ appears before the answer during training, which would favour a held partial result over mixing. Eighteen small transformers were trained from scratch on answer-only $a \pm b - c$: a 6-layer model, a 2-layer block looped three times, and a 2-layer model of the looped one's size, on two-digit numbers and on arithmetic mod 23, three seeds each. Probes with copy ceilings and the operation swap ran every 100 (two-digit) or 50 (mod) training steps. No run showed the signal while the answer was absent. On two-digit numbers, the looped and the size-matched models reach about 99% accuracy with no sign-specific $a \pm b$ at any position or checkpoint ($I \leq 0.034$), and the 6-layer model shows it only after the answer in two of three seeds. On mod 23, all three architectures show it with or after the answer. On a fixed bar of 0.05 the signal crosses later than the answer, but much of that is a threshold effect: the answer signal peaks near 0.97 and I at 0.06–0.26, and measured against each signal's own peak, the two rise within one to three checkpoints of each other in mod 23. Looping made no difference on mod 23; on two-digit numbers the comparison is undefined, since neither the looped nor the size-matched model showed the signal. These checks use the accuracy-based swap.

D. The re-test of Kudo et al. (2026) in detail

Kudo et al. (2026) conclude that language models "have not derived an answer at the moment when input was fed into the model" and compute (sub-)answers while writing the chain of thought. Their evidence is a standard probe for each token position and layer, which before the chain reads the sub-answer at 50.5% and the answer at 17.9%, against 95–100% inside it and a 90% threshold, and patching, in which the input part barely affects the chain. We applied the protocol to their task in a pre-registered study: their two-step problems ($a = k \pm b$, $b = d_1 \pm d_2$, then "a?"), in their format with three worked examples, in Qwen2.5-7B, which writes their chain exactly in 99.8% of problems. We read the states at two positions before the chain, the end of b's equation and the last prompt token. A family varies d_1 and d_2 over every valid pair, in a plus and a minus version; the constant k varies across families, which separates b from the answer. We also ran their short chain format ("$b = 5$, $a = 6$"), in which the chain does not restate b's equation.

Their probe replicates: on single prompts it reads b before the chain at 0.64 (their 50.5%) and a at 0.32 (their 17.9%). The protocol shows what that readability is (Table 5). At both positions and in both formats the operands are combined beyond the copy ceiling: $(d_1 + d_2) \bmod 5$ and $(d_1 - d_2) \bmod 5$ read 0.12–0.38 above it. The original accuracy-based swap called neither position FOLLOWS ($I = 0.00$ and +0.03 against its bar of 0.05), but that swap later failed its positive control (Section 5). Re-scored with the validated share swap (Table 5), both positions are UNCLEAR: the state prefers the difference on both kinds of problem ($D_+ < 0$), with a small

part that follows the operation ($I = +0.020$ and $+0.010$, intervals clear of zero). By the pre-registered rule, nothing places b before the chain. Nor does the chain use such a value. In their format, replacing the whole state at either position with another problem's, from layer 12 on, never changed the b the chain writes in a 40-pair check before the freeze (0 of 40; the main format's patch was not run as a pre-registered test): the chain restates b's equation and computes from that text. In the short format, replacing the state at the end of b's equation changes the written b to the other problem's in 96% of pairs, and so does replacing only the directions of the copied digit (96%); the directions of b with the copy projected out do so in 2%, and the last prompt token feeds that step in none.

Nothing in the protocol's results places the sub-answer before the chain, and the chain computes it from the operands, but the swap cannot tell at these positions. They also show how two weaker tests would have disagreed (their probe, at 0.64, is below their 90% threshold): the operands are combined well beyond the copy ceiling, and clamping b's subspace without removing the copied digit moves the written b in 30% of pairs. Both suggest an early sub-answer. The share swap does not find a code that follows the operation (UNCLEAR), and the copy-controlled clamp finds b's own directions barely move the written b. Their probe also misreads a value that is certainly present (post hoc, descriptive): on the same replication states it reads the copied operand d_1 at 0.59 at the last prompt token, below their 90% threshold, though minimal pairs read it at 1.00 there. It reads the other operand, d_2, at 0.998 at the same token, and both at 1.00 at the end of b's equation, so its rule misses one of the two operands, and only at the end of the question.

Without a chain of thought. Kudo et al. (2026) also have the model answer directly ("$a = 6$"). Their probe then reads b at 69% and a at 41% before the answer, both below their threshold, and they conclude that when the model fails, the sub-answers are unlikely to be represented in its states. Their probe replicates here as well: run as in the chain study (post hoc), it reads b at 0.74 at the token before the answer digit (0.63 within the prompt) and a at 0.32 within the prompt. In this setting the model has to form b, or find a shortcut, without writing it. Qwen2.5-7B answers 69% of random problems like theirs directly (12,000 problems; their 77.8%; our pilot, 67.5%). The gap of about 9 points may reflect differences between our re-implementation of their format and theirs. On our families it answers 76%: 88% when b's operation is a plus and 64% when it is a minus, falling to 38% when both operations subtract. In a second pre-registered study we read the same positions plus a third, the token just before the answer digit. There the answer is itself a confound, since within a family it fixes b. The study therefore also clamps the answer's directions, projects them out of b's along with the copy's, and calibrates a null that mixes the copies with the answer and has no b in it. The operands are again combined beyond the copy ceiling at every position ($+0.14$ to $+0.38$), but under the original accuracy-based swap no position followed the operation (I from 0.00 to $+0.04$, against its bar of 0.05), and mixtures of the copies with the answer, with no b in them, gave as much (up to $+0.06$). Re-scored with the share swap and the answer regressed out, every position is UNCLEAR (Table 5). The clamp gives the same picture. Replacing the whole state with another problem's changes the answer to that problem's in 85% of pairs at the two scorable positions. At the end of b's equation the copied digit's directions alone do this in 83% of pairs. Just before the answer, the answer's own directions do it in 46% and the copy's in 24%. b's directions with both the copy and the answer projected out do it in 1% and 3%; without the projection, b's directions do it in 28% and 35%. Without a chain, then, b's directions beyond the copies and the answer barely matter; since the answer fixes b within a family, the design cannot say whether b is held inside the answer's code. Here too the protocol cannot tell. Their sentence about failures is not contradicted either. Split by whether the model answers correctly, both our operation signal and their probe at first point the other way: on wrongly answered problems I is larger, and their probe reads b at 0.77 against 0.73. But two thirds of the wrong answers come from problems in which both operations subtract, where b reads best. Compared within each pair of operations (post hoc, rule recorded before running), their probe reads b slightly worse on failures (-0.04, 95% CI $[-0.09, +0.01]$).

E. An unwritten value at a distance in detail

Task and models. Prompts are "[pad] $x \circ z$; [k fillers] $w =$" with $x, z, w \in \{0, \ldots, 22\}$, $\circ \in \{+, -\}$, $k = 6$ or 0 (pads keep the length fixed), and answer $y = (T[(x \circ z) \bmod 23] + w) \bmod 23$ for a fixed permutation T. The intermediate $s = (x \circ z) \bmod 23$ is never written, and since w varies the answer does not fix it. The models are 6-layer transformers ($d = 192$) trained from scratch; five seeds reach 0.995–1.000 held-out accuracy, and each run

reproduced, bit for bit, the checkpoint the ground truth was measured on (checked by hash). Negative controls (five seeds, 0.983–0.997) are trained on $y = (T_1[x] + T_2[z] \circ w) \bmod 23$: the same tokens, but $x \circ z$ is never needed.

Ground truth. In pilots, before any protocol statistic, distributed alignment search (Geiger et al., 2024) learned a subspace at each position and layer and swapped it between two prompts. When the swap carries s, the model writes $T[s'] + w$ in 0.78–0.98 of held-out pairs at ";" (layers 3–5) in every model, and in at most 0.07 elsewhere, except 0.13–0.18 at "=" in one model, which is not scored there. Swapping a single operand at ";" succeeds in at most 0.08 of pairs, so ";" holds s itself, not copies of the operands for later use. At layer 0, where the operands are still on their own tokens, the same search swaps each of them in 0.99–1.00 of pairs. The pre-registered rule: a position is positive if s reaches 0.70 with each operand at most 0.15, negative if s stays at most 0.15.

The protocol. The share swap is the one used for Ouro and Kudo (copies x, z, w and the answer regressed out; layers 1–5), with its bar recalibrated on this design from 270 null worlds as in Section 4 (0.35). The clamp follows Section 4 on 1,000 pairs that differ only in z (so in s and the answer, not in x or w), with s's subspace projected off z's copy. It differs in how the codes are found. In a check on a known answer, before the freeze (w's own code at w's token), class means of z carried part of w's code, because w is drawn for each pair, and replacing z's subspace moved w in 40–97% of pairs; the class codes are therefore fitted jointly, by least squares on one-hot codes of s, x, z, w and the operation, and each subspace keeps only the directions above a shuffled-label null. On that check the clamp then gives +0.86 and +0.99 (one model unscorable).

Results. At ";" the clamp gives +0.90, +0.67, +0.60, +0.93, +0.97 over the random subspace at $k = 6$ and +0.85, +0.90, +0.62, +0.91, +0.98 at $k = 0$, every interval's lower end at least +0.54; the whole state moves the answer in 0.82–0.99 of pairs and the random subspace in at most 0.003. It is USED nowhere else (w's token at most +0.09) and in no negative control. The share swap gives I/N 0.78–0.91 at ";" and is FOLLOWS at all 40 scored cells of the five models, fillers, w's token and "=" included, and at none in the negative controls (largest I/N 0.04). One of 270 fresh null worlds is FOLLOWS: a periodic mixture at the weakest mixing strength, $I/N = 0.363$ against 0.35, with a needed share $N = 0.0009$, about fifty times below that of any cell in the trained models. A floor on N would have excluded it, but none was pre-registered, so the swap is not specific by its rule.

Does the earlier clamp transfer? The clamps in Sections 6 and 7 used separate class means. They agree with joint codes when the other inputs are balanced within each class; none of the item sets is exactly balanced (Ouro's by sampling, Kudo's by construction, since b is a function of d_1, d_2 and the operation). Run on the same five models and pairs with separate class means (post hoc), the clamp is USED at ";" in all five at both distances (+0.49 to +0.89 with the copy projected out, +0.75 to +0.93 without it, the form used for Ouro), and USED at no position where s is absent, though also at "=" in the one model where the search found s weakly there (+0.21 and +0.24; joint codes +0.08 and +0.09). There the class means of z's copy also carry part of s's code, which costs the projected clamp some power.

Is s present where the share swap fires? The swap's FOLLOWS at the fillers, w and "=", where DAS and the clamp find no use, could be false positives or could mark a value that is present. An independent check (post hoc) decodes s with a ridge classifier on held-out families, from family-demeaned states with one-hot copies of x, z, w, the operation and the answer regressed out on the training folds (chance 0.04). At the fillers, before w is seen, s reads at 0.90–1.00 in four models and 0.63–0.65 in the fifth, against 0.20–0.31 for random functions of (x, z); in the negative models s reads no better than random functions anywhere (0.07–0.28 against 0.08–0.30). At w and "=", s reads at 0.85–1.00 in four models, but there s is a function of the answer and w, which regressing their one-hot copies does not remove, so these positions are weaker evidence. The value is carried to the fillers and is readable there, but not used: the swap's FOLLOWS there marks presence. The 1 null world in 270 is a true false positive.

Power at a faint signal. In the same five models a further pre-registered study left the forward pass untouched and degraded only the states the clamp's subspaces are fitted on, adding variance drawn once per item until s's share of the remaining variance (the share swap's N, as a population estimate) fell from 0.32–0.47 to 0.1, 0.03, 0.01, 0.005 and 0.002; Ouro's $a + b$ is about 0.008–0.010 on the same scale. The added variance had the covariance of the model's own residual (which crowds s's span at 2.5–5.5 times the average, against 6.5–11.7 for

Ouro's context in the digit's span) or was isotropic, and the subspaces were fitted on all of the fitting half or on 34 items per class, Ouro's ratio of items to dimensions. With separate class means and no copy projection, the form used for Ouro, and s's subspace scored against a random one, the clamp is USED in 5 of 5 models at every dose down to 0.005 in Ouro's regime (+0.73 to +0.90; isotropic, +0.40 to +0.60), and in 3 of 5 at 0.002 (the other two voided by a calibration miss). Ouro's own comparison, against the subspace of the other sign, is weak here at any dose (+0.00 to +0.15 with the model's own covariance), because that subspace moves the answer about as often. The joint-code form of the main test keeps 0–1 directions once the signal is faint or the sample Ouro-sized, and then has no effect. The study degrades the measurement, not the model: a value the model itself codes faintly was not tested.

F. Status of each analysis

"Pre-registered": thresholds and labels were frozen in a commit before the run. "Rule before running": a post-hoc analysis whose reading rule was written down before its result was seen. "Exploratory": decision rule fixed in code before the data, without a frozen pre-registration.

Analysis	Sec.	Status	Outcome
Answer-only accuracy by loop (GSM8K; Huginn check)	A	Pre-registered	passed: +16.5 points over four loops
Re-score with the final scorer	A	Re-run, no test	reproduces the numbers; loop 2–4 gain +3.0 [−1.5, +7.5]
Gain on GSM-Symbolic variants	A	Pre-registered	UNCLEAR: +6.4 points, below the 8.25 bar
Intermediate value at the last prompt token	B.1	Pre-registered	unscorable: sanity check at the majority rate
Copied digits by position; bare arithmetic	B.1	Exploratory	digit read only on its own token; bare operands 0.96–1.00
Minimal pairs; irrelevant number; colour word; value check	B.1	Exploratory, rule in code before data	present at 0.92–1.00 in a shared code; not a magnitude
Standard probe without PCA	B.2	Post hoc, rule before running	still 0.16–0.24
Overlap, structured decoder, learning curve	B.2, B.3	Post hoc, rule before running	swamped: decoder at most 0.21; +0.015 from half to all templates
Crowding by token type; regression on the other numbers' digits	B.3	Post hoc, rule before running	colour crowded too (ratio 0.55: mixed); the other numbers' digits explain none of the unreadability (reported only here); read against own nulls below
Minimal pairs and swamping in Qwen2.5-1.5B	B.2	Pre-registered	replicates: minimal-pair 0.88, standard at most 0.14, decoder at most 0.14
Overlap against its own nulls; MLP probe (both models)	B.2, B.3	Pre-registered (Ouro nulls known before)	overlap 0.83 and 1.01 of the shuffled-label null; MLP at most 0.17
Random-perturbation null (Qwen)	B.3	Pre-registered	generic: word swap 0.89, embedding noise 1.07 of the digit's overlap
Input distance of digits, words and noise (both models)	B.3	Descriptive, no test	digits about half as far apart as random words; word and noise variants equally far apart (Qwen)
Linear response at the varied token (both models)	B.3, B.4	Pre-registered	not tuned to token directions (gain ratio 1.2–1.5; predicted tuned); word swap's effect nonlinear; removing the most amplified directions does not help a probe
Many distinct questions (Qwen, up to 5,249)	B.2	Pre-registered	partial: best probe 0.19; minimal pairs 0.91
Many distinct questions (Ouro, up to 5,249)	B.2	Pre-registered	climbs: best probe 0.36, +0.060 over the last doubling; minimal pairs 0.96

Analysis	Sec.	Status	Outcome
Minimal pairs, swamping and the perturbation null in Qwen2.5-7B and Ouro-2.6B	B.2, B.3	Pre-registered	Qwen replicates (0.85; standard at most 0.27), null generic; Ouro mixed (0.79, below the 0.80 bar), null mixed (word swap quieter)
Many distinct questions to 45,249 (Qwen2.5-7B, Ouro-2.6B)	B.2	Pre-registered; amended before any fit (mean gain added); run split across kernels	Qwen partial: 0.52 at the last layer; Ouro shift-limited: 0.31; minimal pairs 0.86 and 0.79
One reference variant instead of ten (four models)	B.1	Pre-registered	partial in all four: 0.65–0.90 against 0.79–0.96
Computation, Study 1	C.1	Pre-registered	UNCLEAR: +0.118 against a bar of 0.15
Computation, Study 2	C.1	Pre-registered; bar set after Study 1	word problems pass the lowered bar (0.025), not Study 1's 0.15 (mod 10 with the new controls +0.052); bare addition fails
Mod-5 re-scoring of Study 1	C.2	Post hoc, no rule	margin +0.048 [0.030, 0.066]
Sign-flipped twins	C.2	Post hoc, no rule	$a + b$ beats $a - b$ by 0.09 and 0.13
Every combination (31 forms)	C.2	Post hoc, rule before running	$a + b$ first and second: claim weakened
$a - c$ as a partial result	C.2	Post hoc, after the above	not separable from a preference for differences
Target against its answer line	C.2	Post hoc, rule after the estimates were known	6 of 6 lines
Target against other two-number forms	C.2	Post hoc, no rule	$a + b$ leads; $b - c$ not after loop 3
Shown digit and carry-and-borrow baselines	C.2	Post hoc, no rule	margin stays above zero (+0.048)
Template bootstrap; correct-only margins	C.2	Post hoc, no rule	loop-4 intervals not widened; correct-only +0.064
Operation swap (Study 3, Part A)	C.3	Pre-registered	FOLLOWS: I +0.157 [0.141, 0.172] (accuracy swap; see the re-score row)
Patching at P1 after loop 3, same-c pairs	C.3	Pre-registered	NOT-USED; answer patch 0.90
Patching after loop 2; equal-answer pairs	C.3	Pre-registered amendment, after extraction, before analysis	NOT-USED in all three; loop-2 tests weak (answer patch 0.33)
Running total at the end of b's sentence	C.3	Pre-registered	UNCLEAR (accuracy swap, not re-scored): I at most +0.018; patching not run
Copies at P0 by state, Study 3	C.3	Descriptive, rule before running	b 0.77–0.81 after loop 4
Re-test of Kudo et al. (2026): minimal pairs, operation swap, mixing null (Qwen2.5-7B, their Level 3, two chain formats)	7	Pre-registered	accuracy swap, superseded by the re-score row: combined everywhere (+0.12 to +0.38); does not follow at the end of b's equation (I 0.00, mixing); unclear at the last token (I +0.03, beyond mixing, below the 0.05 bar)
Clamp before the chain (short format)	7	Pre-registered; amended before analysis (copy projected out)	descriptive: written b follows the copied digit (96%), b beyond the copy 2%; last token unscorable; their format: no pre-chain state feeds the step (0/40)
Their standard probe on our replication set	7	Pre-registered, descriptive	b 0.64 before the chain (theirs 50.5%), a 0.32 (17.9%)
Their standard probe without a chain	7	Post hoc, descriptive; rule from the chain study, recorded before running	b 0.74 before the answer digit (theirs 69.1%), 0.63 within the prompt; a 0.32 within the prompt (40.8%)

Analysis	Sec.	Status	Outcome
Their probe on wrongly vs correctly answered problems, within operation pairs	7	Post hoc, rule recorded before running	raw split favours failures (0.77 vs 0.73) through the operation mix; within pairs −0.039 [−0.087, +0.008] (unclear), end of b's equation −0.033 [−0.067, −0.000]
Re-test without a chain: minimal pairs, operation swap, answer-mixing null	7	Pre-registered; the frozen code applied the chain study's accuracy gate (0.90) instead of the registered 0.50, corrected after the first run with the statistics unchanged	accuracy swap, superseded by the re-score row: combined at all three positions (+0.14 to +0.38); does not follow at the end of b's equation (I 0.00); unclear before the answer (+0.018) and at the last token (+0.035); answer-mixing null not separable at all three (null up to +0.018, +0.047, +0.057); secondary split: larger I on wrongly answered problems (no label; confounded with the operations)
Clamp without a chain	7	Pre-registered; weak (35–54% of b's directions left)	end of b's equation NOT-USED (+0.005; copy 0.83); before the answer UNCLEAR (+0.025 [0.014, 0.039]; answer 0.46); last token unscorable
Study 3 with carry and borrow in the ceiling	C.3	Pre-registered	P0 explained (−0.013; without them +0.017 [0.007, 0.028]); P1 +0.126
Patching at the end of loop 1; clamping through loops 1–3; patching with the answer ablated	C.4	Pre-registered	loop 1 unscorable (answer patch 0.009); clamp NOT-USED in both pair types (answer clamp 0.914); ablation NOT-USED but without a positive control (output largely scrambled)
Synthetic null for the mixture tests	C.4	Pre-registered	line test NOT-SEPARABLE (null mean margins +0.051, +0.078); null I +0.09, at most +0.14
Positive control: the protocol where Qwen2.5-7B writes b (short chain; eq, last, and one, two and three tokens before b's digit)	5	Pre-registered; code unchanged	fails on the swap: before the digit COMBINED (+0.253), swap UNCLEAR (+0.017), BEYOND-MIXING by 0.0004 (largest null from a mixture that could not be weakened to the observed excess), clamp USED (+0.607 [0.565, 0.646]); swap UNCLEAR at every chain position; clamp unscorable at "=" (whole state 0.10); end of b's equation reproduces the chain study
Their standard probe on the copied operands (replication set)	D	Post hoc, descriptive	d_1 0.59 at the last prompt token (below their 90%), 0.94–1.00 earlier; d_2 at least 0.998 everywhere
Positive control at a distance: clamp at b's written digit, outcome a four tokens later	5	Pre-registered; readings amended after launch, before any output (written-value limit)	USED: +0.806 [0.765, 0.844]; whole state 0.89 from layer 12, 0.90 at every layer; $a = 0 + b$ 0.63
Why the accuracy swap failed (positive-control states)	5	Exploratory	random functions of the operand pair read 0.99–1.00 where b is about to be written; residual-variance shares still follow the operation

Analysis	Sec.	Status	Outcome
Share-based swap: bar from 420 null worlds; fresh controls	5	Pre-registered (bar 0.30 set before the controls)	Qwen2.5-7B a step FOLLOWS (I/N 0.65); Qwen2.5-1.5B b step unscorable (accuracy 0.78 $<$ 0.80); fresh null seeds 0 of 420 FOLLOWS: PARTIAL
Replacement control (Qwen2.5-3B dropped at its accuracy pilot)	5	Pre-registered after the first controls' results, statistic frozen	Mistral-7B b step FOLLOWS (I/N 0.48): share swap VALID
Re-score of the targets with the share swap	6, 7	Pre-registered after validation; old results known; short-chain positions seen during development	Ouro P1 FOLLOWS (I/N 0.63, small share), P0 UNCLEAR; Kudo et al.'s format and answer-only: UNCLEAR everywhere, difference preferred on both kinds of problem
Ouro P1 with the carry and borrow regressed; null on Study 3's design	6	Pre-registered after the re-score (outside review)	FOLLOWS (I/N 0.57; D_+ +0.001, interval just above zero); null 0 of 270, bar 0.25: STANDS
Ouro P1's FOLLOWS against the bar's multiplier and a floor on N	6	Post hoc, on saved results	holds for multipliers up to about 3.8 and floors up to 0.005; rests on D_+, whose interval a 99% level would not clear
Unwritten value at a distance: small transformers trained from scratch (5 seeds; negative controls; ground truth from distributed alignment search)	5	Pre-registered (ground truth and bars from pilots, before any protocol statistic)	clamp USED at ";" 5 of 5 at $k = 6$ and $k = 0$ (+0.60 to +0.98), none in negative controls or where the value is not held; share swap FOLLOWS 5 of 5 (I/N 0.78–0.91) but NOT-SPECIFIC (1 of 270 null worlds, I/N 0.36, needed share 0.0009), and FOLLOWS at every position
Class-mean clamp on the same models	5	Post hoc	USED at ";" 5 of 5 at both distances (+0.49 to +0.89)
Clamp power at Ouro's signal level (fitting data degraded; same models)	6, E	Pre-registered	POWER-AT-OURO-LEVEL: class-mean clamp USED 5 of 5 down to share 0.005 (Ouro about 0.008–0.010); joint-code form fails
Ouro's patches and clamps against the random subspace alone	6	Post hoc, after the power study	loop-3 patch +0.017 [0.008, 0.027] and clamp +0.039 [0.027, 0.051]: UNCLEAR; the other six NOT-USED
Share swap at the end of b's sentence, before c; null on its design	6, C.3	Pre-registered after two outside reviews; the accuracy swap's result there known	UNCLEAR (N 0.002, I +0.000 [0.000, 0.001]); bar 0.35, no null passes; a planted I of 0.002–0.0045 is detected; tail clamp NOT-USED, but the whole-tail patch also hits 0.1% (b's token 0.91)
Training order in small transformers (18 runs, 3 architectures, two-digit and mod 23)	C.6	Pre-registered; timing check against each signal's peak post hoc	no run AHEAD; two-digit looped and size-matched NEVER ($I \leq 0.034$), 6-layer with or after (2/3); mod 23 with or after (all); no looping effect
Normalizing the mid-loop states	C.5	Post hoc, rule before running	alternation real, but general
Profiles across loops; timing	C.2, C.5	Descriptive, no test	answer readable as early as $a + b$